

András Gémes

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Summary

Security researcher with 8 years of experience. [Hands-on experience](#) in malware analysis, reverse engineering and vulnerability research. Certified in [Sec+](#), [CASP+/SecX](#), [CEH](#) and [others](#). LLVM compiler engineering (code obfuscation) and embedded systems background. Open to roles in malware analysis, reverse engineering and vulnerability research.

Work experience

Security Researcher @ HighTec EDV-Systeme GmbH - Budapest, Hungary **Feb 2023 – Present**

- Analyzing Windows, Linux and Android [malware](#), writing [detection rules](#) and config extractors ([Ghidra](#), [Python](#))
- Implementing custom [LLVM-based code obfuscation pass plugins](#) based on malware research
- Evaluating security impact of reported LLVM vulnerabilities as a [member of the LLVM security group](#)
- Hardening the Rust toolchain binaries against reverse engineering

Application Security Engineer @ Knorr-Bremse - Budapest, Hungary **May 2018 – Jan 2023**

- Implemented and evaluated static application security testing across embedded C codebases
- Resolved embedded systems vulnerabilities discovered through code review and AFL++ fuzzing
- Developed embedded C software (memory, RTOS and communication modules)
- Built iOS and Android apps for real-time vehicle data visualization

Technical skills

Reverse engineering (static): DiE, Ghidra, IDA, ILSpy, dnSpy, Binwalk, Apktool, jadx, otool, ipsw

Reverse engineering (dynamic): x64dbg, Sysinternals, LLDB, GDB, Frida, ADB, QEMU, Qiling, VirtualBox, strace

Detection engineering: YARA, Sigma, Suricata, capa

Programming languages: C, C++, Rust, Objective-C, Swift, Python 3, Java, Assembly (ARM64, x86-64), Bash

Vulnerability research: checksec, ROPgadget, AFL++

Network and traffic analysis: Wireshark, Burp Suite, mitmproxy, Zeek, FakeNet-NG, INetSim

Platforms and DevOps tools: Linux (Fedora, Ubuntu), macOS, Windows, Git, Docker, GitHub Actions, Jenkins

Certifications

[CompTIA Security+](#), [CompTIA CASP+/SecurityX](#), [EC-Council CEH](#), [TCM Security PMAT](#), [Invoke RE IMBT](#) and [others](#).

Open source contributions

- [ghidra](#): contributing bug reports and patches to Ghidra, focusing on the BSim, Debugger and FunctionID features
- [threat-detection-rules](#): publishing YARA, Sigma and Suricata rules for analyzed malware families
- [ghidra-scripts](#) | [reversing-scripts](#): developing custom scripts to support reverse engineering
- [phantom-pass](#) | [o-mvll](#): implementing and improving LLVM-based code obfuscation pass plugins
- [rust-arm64](#): writing a Rust book about analyzing Rust-to-ARM64 compilation

Education

MSc in Mechatronics Engineering @ [Budapest University of Technology and Economics](#) **Feb 2016 – June 2018**

BSc in Mechatronics Engineering @ [University of Pannonia](#) **Sept 2012 – Jan 2016**

Continuous education

Currently learning on [Mobile Hacking Lab](#) ([Android](#) and [iOS](#) application security courses).